

CY 552: Hydrogen Generation and Storage

Course Code	CY 552
Course Name	Hydrogen Generation and Storage
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	UG/PG.
Distribution	Elective for B. Tech (3rd year), M. Tech. (Energy Materials), M.Sc Chemistry and Ph. D. students
Approval	9th Senate (May 2016)

Course Contents

- **Production of Hydrogen**

Different methods of hydrogen production, Renewable electrolysis, Steam Methane Reforming, Gasification of coal and other hydrocarbon, partial oxidation of hydrocarbon, Hydrogen from biomass, hydrogen generation from wind energy, water electrolysis, Thermonuclear/other methods and solar energy for hydrogen generation. The course will include at least few Lectures of elaborate study and description about each of the method of hydrogen production. [12 Lectures]

- **Hydrogen Handling and transport**

Major issues regarding the handling of hydrogen and safety requirement .Materials issues : hydrogen embrittlement, Hydrogen leakage and monitoring, hydrogen sensors, Liquid hydrogen: liquefaction costs and low temperature, Pipelines for hydrogen transport-main issue. Few Lectures will be given on these above issues with detailed discussion. Emphasis will be given on the role of materials science in the improvement of the handling of hydrogen. [12 Lectures]

- **Hydrogen Storage**

Hydrogen storage requirements: what is crucial? Metal Hydride requirements for hydrogen storage for mobile application, Onboard hydrogen storage system for light-duty vehicles,Hydrogen storage: materials point of view, Storage tLecturesough surface adsorption: nanocarbon, metal organic framework, Conventional metal hydrides, Light metal hydrides, ClatLecturesates for hydrogen storage. Material Scientists point of view: scope of research on the materials for hydrogen storage, their stability under loading and unloading of hydrogen. [12 Lectures]

- **Hydrogen R & D Opportunities**

Safe handling and safe storage of hydrogen for long term usage, Evaluation of the technique for hydrogen generation, Codes and safety: hydrogen technology standards

Text Books

1. Ram B Gupta (Editor), **Hydrogen Fuel Production, Transport and Storage**, CRC, 2008.

References

1. A.J. Bard and L.R. Faulkner, **Electrochemical Methods**, 2nd Edition, John Wiley and Son, 2001.
2. Marc Koper, Andrzej Wieckowski, (Editors), **Fuel Cell Catalysis: A Surface Science Approach**, Wiley-Interscience, 2009.
3. Marie-Ccile Pera (Editor), Daniel Hissel (Editor), Hamid Gualous (Editor), Christophe Turpin (Editor), **Electrochemical Components** (Electrical Engineering), Wiley-ISTE, 2013